

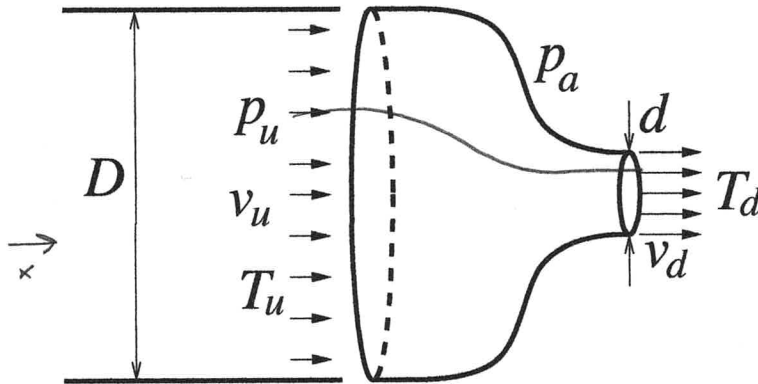
INTRODUCTION TO FLUID MECHANICS

FINAL (Jan 21, 2014)

Problem 1: A liquid of density ρ and specific heat c flows steadily along a circular pipe of diameter D , discharging to the atmosphere, at pressure p_a , through a coaxial contraction that ends with a circular orifice of diameter d . Upstream from the contraction, the velocity, pressure, and temperature have known uniform values v_u , p_u , and T_u . As a function of these data (ρ , c , D , d , p_a , v_u , p_u , T_u), obtain explicit expressions for:

1. The value of the discharge velocity v_d .
2. The force acting on the contraction F , neglecting in the calculation the effect of gravity.
3. The temperature increment $T_d - T_u$. Assume in the calculation that the contraction wall is adiabatic.

Simplify these last two results by further assuming that the effect of viscosity is entirely negligible, so that for this incompressible steady flow Bernoulli's equation can be used to relate the variation of the fluid properties along a given stream line.



$$1) \quad -\rho v_u \pi \frac{D^2}{4} + \rho v_d \pi \frac{d^2}{4} = 0 \Rightarrow \boxed{v_d = \left(\frac{D}{d}\right)^2 v_u}$$

$$2) \quad \vec{F} = - \int_{\Sigma_c} (p - p_a) \vec{n} d\sigma + \int_{\Sigma_c} \vec{\tau} \cdot \vec{n} d\sigma$$

$$+ \rho \pi \frac{D^2}{4} v_u (v_d - v_u) \vec{e}_x = (p_u - p_a) \pi \frac{D^2}{4} \vec{e}_x - \vec{F} \Rightarrow \boxed{\vec{F} = \pi \frac{D^2}{4} \vec{e}_x \left\{ p_u - p_a - \rho v_u^2 \left[\left(\frac{D}{d}\right)^2 - 1 \right] \right\}}$$

$$3) \quad \left[c(T_d - T_u) + \frac{v_d^2}{2} - \frac{v_u^2}{2} \right] \pi \frac{D^2}{4} \rho v_u = + p_u v_u \pi \frac{D^2}{4} - p_a v_d \pi \frac{d^2}{4} \Rightarrow \boxed{T_d - T_u = \frac{1}{c} \left[\left(\frac{p_u}{\rho} + \frac{v_u^2}{2} \right) - \left(\frac{p_a}{\rho} + \frac{v_d^2}{2} \right) \right]}$$

$$4) \quad \frac{p_u}{\rho} + \frac{v_u^2}{2} = \frac{p_a}{\rho} + \frac{v_d^2}{2} \Rightarrow \boxed{\vec{F} = \pi \frac{D^2}{4} \vec{e}_x \rho v_u^2 \left[\left(\frac{D}{d}\right)^2 - 1 \right]^2}$$

$$\boxed{T_d - T_u = 0}$$